

Counting

Basic counting principle - (The sum rule)

If a task can be done with either in one of n_1 ways or in one of n_2 , where none of the set of n_1 ways is the same as any of the n_2 ways, then there are $n_1 + n_2$ ways to do the task.

→ Same time then sum rule

Generalized form of the sum rule:

If the tasks $T_1, T_2, \dots, T_m$ can be done in $n_1, n_2, \dots, n_m$ ways respectively and no two of those tasks can be done at the same time. Then the number of ways to do one of these tasks is $n_1 + n_2 + \dots + n_m$

Ex: A student can choose a project from one of three lists. The three lists contain 23, 15 and 19 possible projects, respectively. How many possible projects are there to choose from?

⇒ $23 + 15 + 19 = 57$ projects to choose from.

Sum rule in terms of sets:

$$|A \cup B| = |A| + |B| \quad |A, B \text{ are disjoint set}|$$

$$\rightarrow |A_1 \cup A_2 \cup \dots \cup A_m| = |A_1| + |A_2| + \dots + |A_m|$$

when $A_i \cap A_j = \emptyset$ for all i, j

The product rule: A procedure can be broken down into a sequence of two tasks. There are n_1 ways to do the 1st task and n_2 ways to do the 2nd task. Then there are $n_1 \cdot n_2$ ways to do the procedure.

→ at different time

Exp: How many bit strings of length seven are there?

→ Since each of the seven bits is either a 0 or a 1, $\therefore 2^7 = 128$ ($2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$)

Generalized form of the product rule:

Suppose that a procedure is carried out by performing the tasks $T_1, T_2, \dots, T_m$. If task T_i can be done in n_i ways after tasks $T_1, T_2, \dots, T_{i-1}$ have been done, then there are $n_1 \cdot n_2 \cdot n_3 \cdot \dots \cdot n_m$ ways to carry out the procedure.

Exp: Suppose a license plate contains two letters followed by three digits with the first digit not zero. How many different license plates can be printed?

→ Each letter can be printed in 26 different ways, the first digit in 9 ways and each of the other two digit in two ways.

$\therefore 26 \cdot 26 \cdot 9 \cdot 10 \cdot 10 = 608400$ different plates can be

Printed.

In term of sets:

$$|A_1 \times A_2 \times A_3 \dots \times A_m| = |A_1| \cdot |A_2| \cdot \dots \cdot |A_m|$$

Combining the sum and product rule:

Many counting problem cannot be solved using sum rule or just product rule. Complicated counting problem can be solved using both of these rules.

Exp: Suppose statements labels in a programming language can be either a single letter or a letter followed by a digit. Find the number of possible labels.

(Solution: use product rule,

$$26 + 26 \cdot 10 = 286$$

Counting password:

Exp: Each user on a computer system has a password, which is six to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at list one digit. How many possible ~~character~~ passwords are there?

Solution: Let p be the number of passwords and P_6, P_7 and P_8 be the passwords of length 6, 7, 8.

By sum rule, $p = P_6 + P_7 + P_8$

26 letters 10 digit	} 36

$$P_6 = 36^6 - 26^6$$

$$= 2,176,782,336 - 308,915,776$$

$$= 1,867,866,560$$

$$P_7 = 36^7 - 26^7 = 78,364,164,2096 - 8,031,810,176$$

$$= 70,332,253,920$$

$$P_8 = 36^8 - 26^8 = 2,821,109,907,456 - 208,827,064,576$$

$$= 2,612,282,842,880$$

$$P = P_6 + P_7 + P_8$$

$$= 2,684,483,063,360$$

Subtraction rule/ Inclusion - Exclusion principle:

If a task can be done either in one of n_1 ways or in one of n_2 ways, then the total number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways.

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Generalized principle of Inclusion - Exclusion;

The principle of inclusion - exclusion can be generalized to find the number of ways to do one of n different tasks or equivalently to find the number of elements in the union of sets.

Let $A_1, A_2, \dots, A_n$ be finite sets. Then,

$$|A_1 \cup A_2 \cup \dots \cup A_n| = \sum |A_i| - \sum |A_i \cap A_j| + \sum |A_i \cap A_j \cap A_k| \\ - \dots + (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n|$$

Exp: Give a formula for the number of elements in the union of four sets.

$\Rightarrow$ The inclusion - exclusion principle shows that

$$|A_1 \cup A_2 \cup \dots \cup A_n| = |A_1| + |A_2| + |A_3| + |A_4| - |A_1 \cap A_2| - |A_1 \cap A_3| \\ - |A_1 \cap A_4| - |A_2 \cap A_3| - |A_2 \cap A_4| - |A_3 \cap A_4| + |A_1 \cap A_2 \cap A_3| \\ + |A_1 \cap A_2 \cap A_4| + |A_1 \cap A_3 \cap A_4| + |A_2 \cap A_3 \cap A_4| - |A_1 \cap A_2 \cap A_3 \cap A_4|.$$

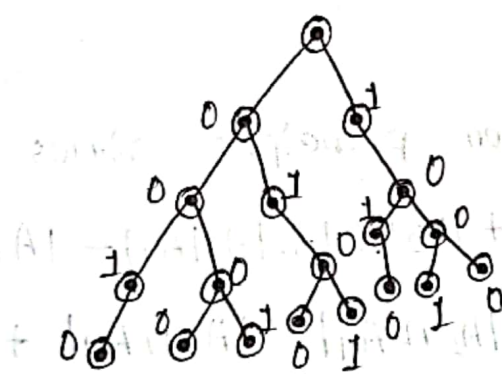
$\therefore$ This formula contains 15 different items

$\therefore$ i.e., $2^4 - 1$

Tree diagrams: Many counting problems can solve through the use of tree diagrams, where a branch represents a possible choice and the leaves represent possible outcomes.

Exp: How many bit strings of length four do not have two consecutive 1's.

⇒ The following tree diagram displays all bit strings of length four without two consecutive.



1 হলে দুই 0 এর
দানের টা,
আর 0 হলে দানসে
1 and 0

0000 0001 0010 0011 0100 0101 0110 0111 1000 1001 1010 1011 1100 1101 1110 1111

Pigeonhole principle: If k is a positive integer and $k+1$ objects are placed into k boxes, then at least one box contains two or more objects.

~~Statements: The pigeonhole principle states that if we must put $N+1$ or more~~

$\Rightarrow$ If there are more pigeons than pigeonholes then there must be at least one pigeonhole with at least two pigeons in it.

Corollary 1: A function f from a set with $k+1$ elements to a set with k elements is not one to one.

Proof: $\Rightarrow$ Create a box for each element y in the codomain of f

$$\Rightarrow f(x) = y$$

$\Rightarrow$ there are $k+1$ elements and only k boxes, at least one box has two or more elements.

$\therefore f$ can't be one to one.

- Pigeonhole principle $\rightarrow$ Another name is Dirichlet's drawer Principle

The Generalized pigeonhole principle:

If N objects are placed into k boxes, then there is at least one box containing at least $\lceil N/k \rceil$ objects.

$\Rightarrow$ If none of boxes contains more than $\lceil N/k \rceil - 1$ objects then the total number of object is at most,

$$k \left(\lceil \frac{N}{k} \rceil - 1 \right) < k \left(\left(\frac{N}{k} + 1 \right) - 1 \right) = N,$$

Permutations: A permutation is a set of distinct object is an ordered arrangement of these objects, An ordered arrangement of r elements of a set is called an r -permutations.

Formula:

Theorem 1 - If n is a positive integer and r is an integer $1 \leq r \leq n$ then there are

$$P(n, r) = n(n-1)(n-2) \dots (n-r+1)$$

Proof: Use the product rule, The first element can be chosen in ' n ' ways. The 2nd in $n-1$ ways. and so on until there are $(n - (r-1))$

ways to choose the last element:

If n and r are integers with $1 \leq r \leq n$ then

$$P(n, r) = \frac{n!}{(n-r)!}$$

Permutation with Repetition:

Theorem : 2

The number of r -permutations of a set of n objects with repetition allowed is n^r .

With indistinguishable objects:

Theorem : 03

The number of diff. permutations of n objects where there are n_1 indistinguishable objects of type 1, n_2 indistinguishable objects of type 2 and n_k indistinguishable objects of type k is:

$$\frac{n!}{n_1! n_2! \dots n_k!}$$

Combination: An n -combinations of element of a set is an unordered selection of n elements from the set. Thus an n combinations is simply a subset of the set with ' n ' elements.

→ n -combination of a set with n -distinct element is denoted by $c(n, n) / c\left(\begin{smallmatrix} n \\ n \end{smallmatrix}\right)$.

Theorem: 1

The number of n -combinations of a set with n elements where $n > n > 0$ equals,

$$c(n, n) = \frac{n!}{(n-n)!n!}$$

Proof: By the product rule,

$$P(n, n) = c(n, n) \cdot P(n, n)$$

$$c(n, n) = \frac{P(n, n)}{P(n, n)}$$

$$= \frac{n! / (n-n)!}{n! / (n-n)!}$$

$$= \frac{n!}{(n-n)!n!}$$

Corollary 2:

Let n and r be the nonnegative integers with $r \leq n$. Then $C(n, r) = C(n, n-r)$

Proof: From theorem, $C(n, r) = \frac{n!}{(n-r)!r!}$

$$\text{and, } C(n, n-r) = \frac{n!}{(n-r)! [n-(n-r)]!}$$
$$= \frac{n!}{(n-r)!r!}$$

Hence, $C(n, r) = C(n, n-r)$

Combination with repetition:

Theorem: There are $C(n+r-1, r)$ r -combinations from a set with n elements when repetition of elements is allowed.

Formula:

Type	Repetition	Formula
r -permutations	NO	$\frac{n!}{(n-r)!}$
r -permutations Combinations	NO	$\frac{n!}{r!(n-r)!}$
r -permutations	Yes	n^r
r -combinations	Yes	$\frac{(n+r-1)!}{r!(n-1)!}$

Binomial Expression: A binomial expression is the sum of two terms, such as $(x+y)$ [more generally, these terms can be products of constants and variables]

Pascal's Identity:

If 'n' and 'k' are integers with $n \geq k \geq 0$, then
$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$$

			1			
		1		1		
	1		2		1	
1		3		3		1
1	4		6		4	1
1	5	10		10	5	1
1	6	15	20	15	6	1

The n^{th} row in the triangle consists of the binomial coefficients $\binom{n}{k}$; $k = 0, 1, 2, \dots, n$

Theorem: Let n be a positive integer, Then

$$\sum \binom{n}{k} = 2^n$$

Theorem: Let m, n and r be non negative integers with r not exceeding either 'm' or 'n'. Then
$$\binom{m+n}{r} = \sum \binom{m}{r-k} \binom{n}{k}$$

Binomial theorem: Let 'x' and 'y' be variables and 'n' a nonnegative integer.

Then,

$$(x+y)^n = \sum_{j=0}^n \binom{n}{j} x^{n-j} y^j = \binom{n}{0} x^n + \binom{n}{1} x^{n-1} y + \dots + \binom{n}{n-1} x y^{n-1} + \binom{n}{n} y^n$$

Exp: What is the coefficient of $x^{12} y^{13}$ in the expansion of $(2x-3y)^{25}$?

$$\Rightarrow {}^C(25, 13) 2^{12} (-3)^{13} = - (25! / 13! 12!) 2^{12} \cdot 3^{13}$$

⑦ Theorem: Let n be a positive integer,

$$\text{Then } \sum (-1)^k {}^C(n, k) = 0$$